$$A = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \qquad C = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \qquad D = \begin{pmatrix} 1 & 2 \end{pmatrix}$$

$$\underset{2\times 3}{B} \quad \underset{3\times 1}{C} = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix}\begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}$$

product 2×1

$$= 1\begin{pmatrix} 1 \\ 0 \end{pmatrix} + 0\begin{pmatrix} 0 \\ 1 \end{pmatrix} + 3\begin{pmatrix} 3 \\ 0 \end{pmatrix} = \begin{pmatrix} 10 \\ 0 \end{pmatrix}.$$

$$A \quad B = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \qquad \begin{pmatrix} -1 \\ 0 \end{pmatrix} \quad 1\begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

$$= \left(1\begin{pmatrix} -1 \\ 1 \end{pmatrix} + 0\begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad 0\begin{pmatrix} -1 \\ 1 \end{pmatrix} + 1\begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad 3\begin{pmatrix} -1 \\ 1 \end{pmatrix} + 0\begin{pmatrix} 0 \\ 2 \end{pmatrix}. \right)$$

$$= \begin{pmatrix} -1 & 0 & -3 \\ 1 & 2 & 3 \end{pmatrix}$$

$$A = \begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix} \qquad B = \begin{pmatrix} 1 & 0 & 3 \\ 0 & 1 & 0 \end{pmatrix} \qquad C = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix} \qquad D = \begin{pmatrix} 1 & 2 \end{pmatrix}$$

Can $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ be written as a linear combination of $\begin{pmatrix} -1 \\ 1 \end{pmatrix}$ and $\begin{pmatrix} 0 \\ 2 \end{pmatrix}$?

Does $\begin{pmatrix} -1 & 0 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ have a solution?

Does the system of linear eqns $\begin{cases} -x_1 + 0x_2 = 1 \\ x_1 + 2x_2 = 2 \end{cases}$ have a solution?